

SEEDLINE · PROBLEM STATEMENT

SeedLine problem statement

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (problem, users, context, constraints, out of scope, prior work with sources)
0.1	2026-09-24	Amish Chadha	Initial scaffold

SeedLine problem statement

Hand-seeding is slow and uneven, and precision planters are too costly for small plots. The gap is a push seeder that places one seed at a time at a set spacing, costs about the same as the cheapest plate seeder, and can be adapted to any crop by printing a new metering plate instead of buying one.

■ The problem

Most of the world's farms are small. About 84 % of the world's roughly 570 million farms are under 2 ha ([Our World in Data](#), from [Lowder et al.](#)). On these plots and on market gardens, seed is still sown by hand with a hoe, a stick or a jab planter, or dribbled from a cheap plate seeder.

Three problems follow:

1. **Labor.** Hand planting with a hoe has been estimated at about 56 h per hectare for one worker in southern Africa, and a jab planter cuts that to about a third ([Baudron et al., cited in a CSBE jab planter study](#)). Planting happens in a short window after the first rains, so slow planting delays part of the crop.
2. **Uneven spacing.** Hand-dropped seed lands in clumps and gaps. Uneven stands cost yield: Purdue trials on maize found a loss of about 2.2 to 2.5 bu/acre (about 140 to 160 kg/ha) for each inch (25 mm) increase in the standard deviation of plant-to-plant spacing ([Nielsen, Purdue University](#)). Clumped vegetable seed also has to be thinned by hand.
3. **Cost and fit of existing tools.** Low-cost plate seeders such as the Earthway 1001-B (about \$187 with six plates) pick up several seeds at a time, so crops usually need thinning, while singulating roller seeders such as the Jang JP-1 cost about \$499 before rollers at about \$25 each ([University of Minnesota Extension, 2024](#)). Users also report that the Earthway is inaccurate with small seed and has no in-row spacing adjustment ([Paperpot Co.](#)). A farmer whose crop or seed size does not match a stock plate has no easy way to get one.

SeedLine is a single-row push seeder whose ground wheel drives a vertical metering plate through a chain. The plate is 3D printed for each crop and seed size, so a farmer, a cooperative or a local maker with a printer can make a plate for any seed. An optional row-marker kit sets the spacing between rows.

■ Users and context

User	Need	Context
Smallholder farmer	Plant maize, beans, sorghum or groundnuts at the right spacing in the planting window, with less labor and less seed	Plots of 0.2 to 2 ha, hand-tilled or ox-ploughed soil, rain-fed, often one worker
Market gardener	Sow vegetables in beds without thinning, at low cost	Beds of 0.75 to 1.2 m, fine tilled soil, 3 to 6 rows per bed
Cooperative or agro-dealer with a printer	Print plates on demand for members' crops and varieties; lend or rent seeders	Rural town, one FDM printer, basic tools
Local welder or maker	Build and repair the frame from common steel tube and bicycle parts	Workshop with a welder or drill, bicycle spares available
Extension worker or researcher	Recommend spacings and check stands	Field demonstrations, trial plots

Operating environment

- **Soil:** tilled loam to sandy loam, with clods, crop residue and stones up to about 50 mm; seedbed quality varies widely on hand-tilled plots.
- **Seed:** from maize and beans (8 to 15 mm) down to vegetable seed of 1 to 2 mm; some seed is treated with fungicide or insecticide.
- **Climate:** 10 to 40 °C, strong sun, dust; rain and mud at planting time.
- **Supply chain:** steel tube, bicycle chain and sprockets, bearings and wheelbarrow or trolley wheels are common in regional towns; 3D printers are uncommon in villages but increasingly present in towns, universities and maker spaces.

■ Constraints

- Garage-buildable prototype, about \$400 USD (`project.yaml`).
- A replicated base seeder (without the marker kit) should cost no more than a low-cost plate seeder, about \$200 in parts.
- Metering plates printable in PLA, PETG or ASA on a common desktop FDM printer, without supports.
- Frame and drive from common steel sections, bicycle chain parts and bearings that local workshops stock.
- Pushed by one person at walking speed on a tilled seedbed.
- No engine, battery or electronics in the base design.

■ Out of scope

- Tractor or animal-drawn planters.
- Fertilizer placement (a later option, not part of this concept).
- Transplanting, seed-tape or paper-pot systems.
- Seed treatment, seed supply or agronomic advice beyond the plate spacing tables.
- Vacuum or air-assisted metering.

■ Prior work

- **Commercial push seeders.** The Earthway plate seeder and the Jang roller seeder set the price and performance range for push seeders ([UMN Extension](#); [Paperpot Co.](#)). Both use interchangeable plates or rollers bought from the maker.
- **Open 3D-printed seeders.** The Hour Farm published a hackable, 3D-printed precision seeder with interchangeable printed seed discs on FarmHack in 2019 under CC BY 4.0 ([FarmHack](#)). It shows that printed discs work for vegetable seed at bed scale.
- **Printed parts in precision metering.** A Norfolk farmer printed singulator discs and sprockets for a six-row vacuum maize planter, reporting about 99 % singulation and no visible wear after 70 ha ([3D Printing Industry, 2019](#)). This is a tractor planter, but it supports the durability of printed metering parts.
- **Jab planters.** Hand jab planters are promoted for conservation agriculture in Africa and cut planting time to about a third of hoe planting ([CSBE study](#); [FAO jab planter user manual](#)). They place seed hill by hill, so spacing depends on the operator.
- **Research push planters.** Many university studies build and test manual single-row planters with plate or cell-wheel metering (for example, [a manually operated planter for different seeds](#)). Their field capacity and seed placement data will be reviewed at TRL 3 to check the estimates in SDL-PRC-001.

■ Open questions

- Which users first: smallholder field crops (maize and beans at 0.75 m rows) or market-garden vegetables? This changes the wheel, opener and plate range. Proposed, awaiting Amish (see SDL-PRC-001).
- Which partner and region for co-design and field trials? Proposed, awaiting Amish.
- Who prints plates in practice: the farmer, a cooperative, an agro-dealer or a central maker who mails them?
- How common is treated seed among target users, and what handling rules apply?
- Is the row marker enough, or do users want two or three units ganged on one frame?

■ User research and co-design

Requirements must come from the people who will use the seeder.

- [] Identify a local partner organization (farmer cooperative, extension service, NGO or university)
- [] Run co-design sessions with intended users; record who, where and what was learned
- [] Validate crop list, spacing, soil and cost assumptions in the field
- [] Revise requirements (REQ) from findings before freezing the design